

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 35%

QUESTIONS: 05

Total Marks:50

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding (2.5)	Constraint Analysis (2.5)	Solution Development (2.5)	Application of Concepts (1.5)	Justification (1)	Total(10)
Weightage	25%	25%	25%	15%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I G Madhulatha(192311295)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G Madhulatha

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization

Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving (6 Marks) (BL4–BL5) – Reframed with New Scenarios

Q1. Dataset: Hospital Patient Records

PID	Symptoms Observed
P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Question:

Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include "Fever". Identify the valid association rules and justify why they satisfy the given constraints. (BL4 – 1.2 Marks)

Q2. Dataset: Online Learning Platform

Student ID	Courses Enrolled
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Question:

Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain at least two courses. Explain how the constraint helps in pruning candidate itemsets during mining. (BL4 – 1.2 Marks)

Q3. Dataset: Library Borrowing Records

Transaction ID	Books Borrowed
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Question:

Apply the Apriori algorithm with the constraint that the maximum itemset size is 3. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation. (BL5 – 1.2 Marks)

Q4. Dataset: University Course Hierarchy

Student ID	Courses Taken
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Question:

Apply multilevel association rule mining with the constraint that rules should be generated only at the "Computer Science" level. Explain how the course hierarchy affects the discovery of association rules. (BL5 – 1.2 Marks)

Q5. Dataset: Insurance Customer Records

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question:

Apply multidimensional association rule mining with the constraint Age Group = "Young". Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies. (BL4 – 1.2 Marks)

CONSTRAINT-BASED PROBLEM SOLVING

Q1. Dataset: Hospital Patient Records

Given Data

P1 – Fever, Cough

P2 – Fever, Headache

P3 – Cough, Headache

P4 – Fever, Cough, Headache

P5 – Fever, Fatigue

Minimum Support = 50%

Minimum Confidence = 70%

Step 1: Calculate Minimum Support Count

Number of transactions = 5

Minimum support count = $50\% \times 5$

= 2.5

= 3 transactions

Therefore, an itemset must occur in at least 3 patients.

Step 2: Frequent 1-Itemsets

Fever occurs in P1, P2, P4 and P5.

Support of Fever = $4/5 \times 100 = 80\%$

Cough occurs in P1, P3 and P4.

Support of Cough = $3/5 \times 100 = 60\%$

Headache occurs in P2, P3 and P4.

Support of Headache = $3/5 \times 100 = 60\%$

Fatigue occurs only in P5.

Support of Fatigue = $1/5 \times 100 = 20\%$

Therefore, the frequent 1-itemsets are:

{Fever} – 80%

{Cough} – 60%

{Headache} – 60%

Fatigue is not frequent.

Step 3: Frequent 2-Itemsets

{Fever, Cough} occurs in P1 and P4.

Support = $2/5 \times 100 = 40\%$

{Fever, Headache} occurs in P2 and P4.

Support = $2/5 \times 100 = 40\%$

{Cough, Headache} occurs in P3 and P4.

Support = $2/5 \times 100 = 40\%$

All 2-itemsets have only 40% support, which is less than the required 50%.

Therefore, there are no frequent 2-itemsets.

Association Rules

The constraint says that every rule must include Fever.

Possible rules are:

Fever $\rightarrow$ Cough

Confidence = $\text{Support}(\text{Fever, Cough}) / \text{Support}(\text{Fever})$

= $2/4 \times 100$

= 50%

Cough $\rightarrow$ Fever

Confidence = $2/3 \times 100$

= 66.67%

Fever $\rightarrow$ Headache

Confidence = $2/4 \times 100$

= 50%

Headache $\rightarrow$ Fever

Confidence = $2/3 \times 100$

= 66.67%

None of the rules satisfies the minimum confidence of 70%.

Final Answer

No valid association rule exists under the given constraints.

Justification: All Fever-containing 2-itemsets have only 40% support, which is below the required minimum support of 50%. Also, none of the possible rules reaches the required confidence of 70%.

Q2. Dataset: Online Learning Platform

Given Data

S1 – Python, SQL

S2 – Python, Data Science

S3 – SQL, Machine Learning

S4 – Python, SQL, Data Science

S5 – Python, Machine Learning

Minimum Support = 40%

Step 1: Calculate Minimum Support Count

Number of students = 5

Minimum support count = $40\% \times 5$

= 2 transactions

Therefore, an itemset must occur in at least 2 students.

Step 2: Frequent 1-Itemsets

Python occurs in S1, S2, S4 and S5.

Support = $4/5 \times 100 = 80\%$

SQL occurs in S1, S3 and S4.

Support = $3/5 \times 100 = 60\%$

Data Science occurs in S2 and S4.

Support = $2/5 \times 100 = 40\%$

Machine Learning occurs in S3 and S5.

Support = $2/5 \times 100 = 40\%$

Therefore, all four courses are frequent:

{Python} – 80%

{SQL} – 60%

{Data Science} – 40%

{Machine Learning} – 40%

Step 3: Frequent 2-Itemsets

{Python, SQL} occurs in S1 and S4.

Support = $2/5 \times 100 = 40\%$

{Python, Data Science} occurs in S2 and S4.

Support = $2/5 \times 100 = 40\%$

{Python, Machine Learning} occurs only in S5.

Support = $1/5 \times 100 = 20\%$

{SQL, Data Science} occurs only in S4.

Support = $1/5 \times 100 = 20\%$

{SQL, Machine Learning} occurs only in S3.

Support = $1/5 \times 100 = 20\%$

{Data Science, Machine Learning} does not occur.

Support = 0%

Therefore, the frequent itemsets containing at least two courses are:

{Python, SQL} – 40%

{Python, Data Science} – 40%

Constraint-Based Pruning

The constraint states that every itemset must contain at least two courses.

Therefore, single-course itemsets are not considered as final answers. The algorithm focuses on combinations containing two or more courses.

Final Answer

Frequent itemsets satisfying the constraint are:

{Python, SQL} – 40%

{Python, Data Science} – 40%

The constraint reduces unnecessary candidates and makes the mining process more efficient.

Q3. Dataset: Library Borrowing Records

Given Data

T1 – Data Mining, Python Programming

T2 – Data Mining, Database Systems

T3 – Python Programming, Database Systems

T4 – Data Mining, Python Programming, Database Systems

T5 – Data Mining, Artificial Intelligence

Maximum Itemset Size = 3

Note: The question does not specify a minimum support value. Therefore, the support counts and percentages are shown below. If the minimum support is 40%, the frequent itemsets are identified accordingly.

Step 1: 1-Itemsets

Data Mining occurs in T1, T2, T4 and T5.

Support = $4/5 \times 100 = 80\%$

Python Programming occurs in T1, T3 and T4.

Support = $3/5 \times 100 = 60\%$

Database Systems occurs in T2, T3 and T4.

Support = $3/5 \times 100 = 60\%$

Artificial Intelligence occurs only in T5.

Support = $1/5 \times 100 = 20\%$

Therefore:

{Data Mining} – 80%

{Python Programming} – 60%

{Database Systems} – 60%

{Artificial Intelligence} – 20%

Step 2: 2-Itemsets

{Data Mining, Python Programming} occurs in T1 and T4.

Support = $2/5 \times 100 = 40\%$

{Data Mining, Database Systems} occurs in T2 and T4.

Support = $2/5 \times 100 = 40\%$

{Python Programming, Database Systems} occurs in T3 and T4.

Support = $2/5 \times 100 = 40\%$

{Data Mining, Artificial Intelligence} occurs in T5.

Support = $1/5 \times 100 = 20\%$

{Python Programming, Artificial Intelligence} occurs in no transaction.

Support = 0%

{Database Systems, Artificial Intelligence} occurs in no transaction.

Support = 0%

Step 3: 3-Itemsets

{Data Mining, Python Programming, Database Systems} occurs only in T4.

Support = $1/5 \times 100 = 20\%$

Therefore, if minimum support is 40%, this 3-itemset is not frequent.

Frequent Itemsets at 40% Minimum Support

1-itemsets:

{Data Mining} – 80%

{Python Programming} – 60%

{Database Systems} – 60%

2-itemsets:

{Data Mining, Python Programming} – 40%

{Data Mining, Database Systems} – 40%

{Python Programming, Database Systems} – 40%

Effect of Maximum Itemset Size Constraint

The maximum itemset size is 3.

Therefore, Apriori generates only 1-itemsets, 2-itemsets and 3-itemsets.

Itemsets containing 4 or more books are never generated.

This reduces the candidate search space and improves the efficiency of the Apriori algorithm.

Q4. Dataset: University Course Hierarchy

Given Data

S1 – Engineering, Computer Science, Machine Learning

S2 – Engineering, Information Technology

S3 – Engineering, Computer Science

S4 – Engineering, Computer Science, Machine Learning

S5 – Engineering, Information Technology

Course Frequencies

Engineering occurs in S1, S2, S3, S4 and S5.

Support = $5/5 \times 100 = 100\%$

Computer Science occurs in S1, S3 and S4.

Support = $3/5 \times 100 = 60\%$

Machine Learning occurs in S1 and S4.

Support = $2/5 \times 100 = 40\%$

Information Technology occurs in S2 and S5.

Support = $2/5 \times 100 = 40\%$

Association Patterns

Engineering → Computer Science

Support = $3/5 \times 100$

= 60%

Computer Science → Machine Learning

Both Computer Science and Machine Learning occur together in S1 and S4.

Support = $2/5 \times 100$

= 40%

Engineering → Information Technology

This combination occurs in S2 and S5.

Support = $2/5 \times 100$

= 40%

Effect of Course-Level Constraint

The constraint specifies that rules should be generated only at the selected course level.

Therefore, the mining algorithm does not generate unnecessary rules from other hierarchy levels.

It focuses on meaningful course associations such as:

Engineering → Computer Science

Engineering → Information Technology

Computer Science → Machine Learning

This reduces the search space and produces more relevant association rules.

Final Answer

The course-level constraint controls the hierarchy level at which association rules are generated. It improves efficiency by eliminating rules from unwanted levels and helps discover meaningful relationships between university courses.

Q5. Dataset: Insurance Customer Records**Given Data**

C1 – Young, High, Health Insurance

C2 – Middle, Medium, Vehicle Insurance

C3 – Young, Low, Travel Insurance

C4 – Senior, High, Health Insurance

C5 – Young, Medium, Vehicle Insurance

Constraint:

Age Group = "Young"

Step 1: Apply the Constraint

Only customers belonging to the Young age group are considered.

The selected customers are:

C1 – Young, High, Health Insurance

C3 – Young, Low, Travel Insurance

C5 – Young, Medium, Vehicle Insurance

Therefore, the constrained dataset contains 3 customers.

Step 2: Identify Patterns

Pattern 1:

Young + High Income → Health Insurance

This pattern occurs for C1.

Support among Young customers:

$$1/3 \times 100 = 33.33\%$$

Pattern 2:

Young + Low Income → Travel Insurance

This pattern occurs for C3.

Support among Young customers:

$$1/3 \times 100 = 33.33\%$$

Pattern 3:

Young + Medium Income → Vehicle Insurance

This pattern occurs for C5.

Support among Young customers:

$$1/3 \times 100 = 33.33\%$$

Meaningful Patterns

Young + High Income → Health Insurance

Young + Low Income → Travel Insurance

Young + Medium Income → Vehicle Insurance

Marketing Interpretation

These patterns can help an insurance company target young customers according to their income level.

Young customers with high income can be targeted with Health Insurance.

Young customers with low income can be targeted with Travel Insurance.

Young customers with medium income can be targeted with Vehicle Insurance.

Therefore, the constraint Age Group = "Young" reduces the dataset and focuses the mining process on the desired customer segment.

This makes the discovered patterns useful for targeted marketing strategies.

Final Answer

The constraint-based mining identifies meaningful relationships between income level and insurance policy among young customers. It reduces the search space and allows the insurance company to create targeted marketing campaigns for young customers based on their income level.